



Ableton Tools ▾

Learn Ableton

My AccountAUD



(https://elphnt.io) Newsletter

Info ▾

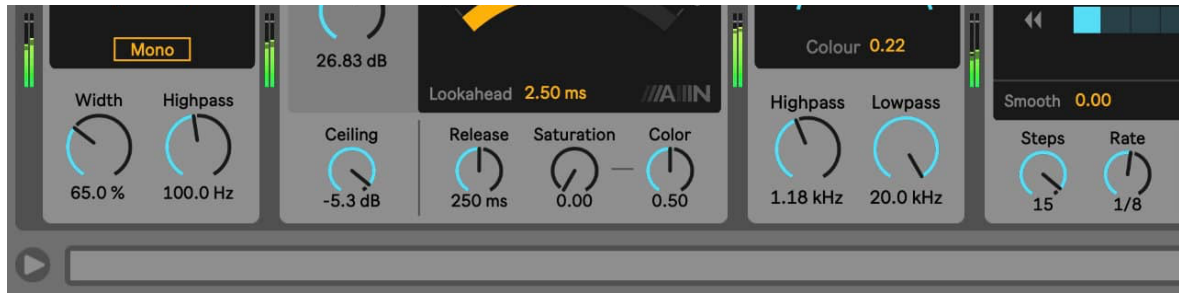
Learn How To Build Your Own Max For Live Devices

MAX FOR LIVE, TUTORIAL • 19 JANUARY 2024

You've heard about Max For Live and some of the amazing things you can do with it (<https://elphnt.io/what-is-maxforlive/>). Maybe you've even used a few Max For Live devices (<https://elphnt.io/max-for-live-curated-collection/>). Now you want to take things to the next level and build your own custom instruments and effects to make the sounds you've always dreamed of.

This guide is here to help you do just that.





A selection of Max For Live devices

Getting started programming in Max can be a little daunting, so I hope to simplify the process as much as possible. The resources that exist for learning Max tend to be a little sporadic and hard to follow, so my intention is also to provide a useful, linear path you can follow to get to making your own tools as quickly and painlessly as possible.

Max and Max For Live are two related, but slightly different things. I'll explain the difference between them shortly, but for now, I'll use the shorthand Max to refer to both.

Why Learn Max?

Before getting started it's probably worth understanding why you might want to build Max devices in the first place.

Build Your Own Instruments & FX

The first answer is obvious – you want to build custom instruments and effects but don't want to go through the daunting task of having to learn something like C++ and how to build VST plugins.

Once you've mastered some sound design basics and know a little about how sound works, Max is the next step in going even deeper into sound design. It's by far the most approachable and easy-to-learn way of building custom audio tools and can even serve as a great stepping stone into building VST plugins if you want to go that route.



A custom analog-style EQ built in Max For Live

Max Is Fun!

There's something supremely rewarding about solving problems and Max is all about that; presenting yourself with a particular audio-based problem, then building a solution for it. There's nothing quite like the feeling of spending hours working on a complex patch, firing it up and hearing that first successful bit of processed sound.

Improve Your Sound Design

Max will also deepen your knowledge of sound design and DSP (digital signal processing). If you want to truly understand how things like delays, reverbs and all manner of audio processes work then Max will get you deep into the guts of these tools, giving you a broader understanding of how they work and how to master them.

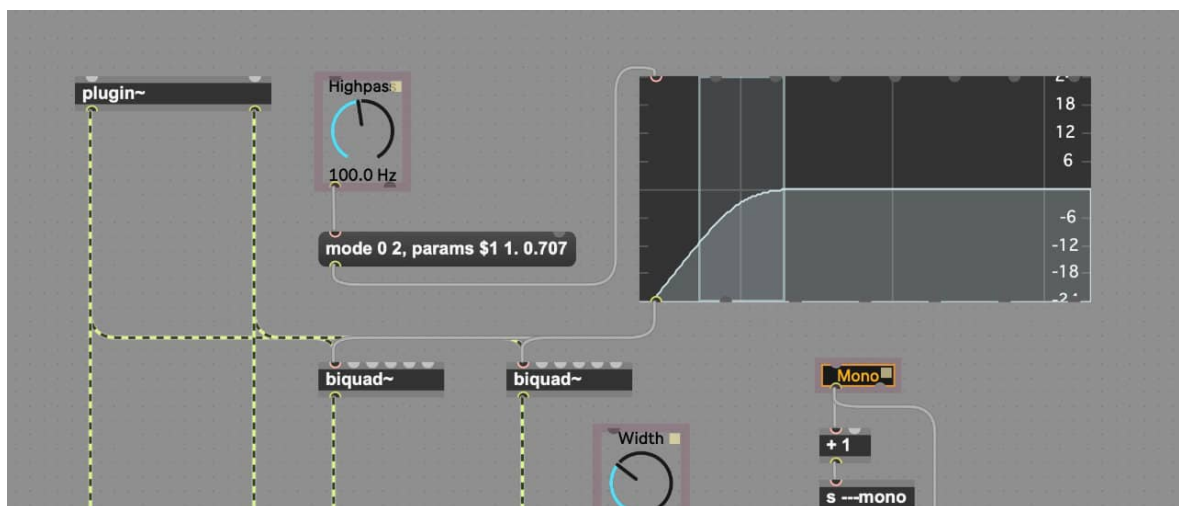
Earn A Living

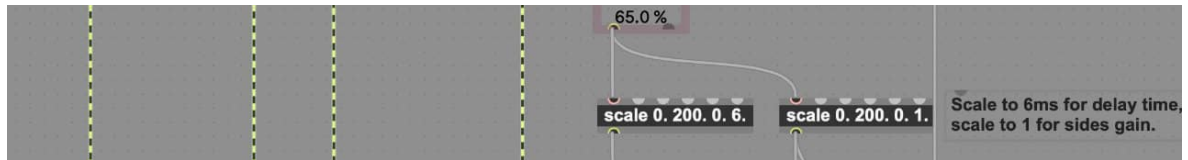
Building Max devices can be a career too, if you want to go that route. I've been building and selling Max devices for many years (<https://elphnt.io/products/ableton-live-packs/>) and a large part of my income comes from that.

Despite this, I still build plenty of Max devices for my own use and Max continues to be a place of personal sound design exploration for me.

What Is Max/Max For Live?

Max is a visual programming language. Instead of writing lines of code, Max works more like a modular synth.





Inside a typical Max For Live patch

Max has hundreds of 'objects' that each do something particular to your signal and you connect these objects using patch cables.

It's a lot more accessible and immediate than typing out lines of code, though you can still write code in a conventional way inside of Max using JavaScript and GenExpr (Max's own C-like language) if you'd like.

Max vs Max For Live

It's worth distinguishing Max and Max For Live.

Max is the broader application and programming environment. Max For Live is simply an implementation of Max that runs inside Ableton Live and contains objects meant for use inside of Live.

The original Max application has been around for a long time (the Max Wikipedia article (https://en.wikipedia.org/wiki/Max_%28software%29) is a good read for a bit of Max history). Max For Live was created as a partnership between Ableton and Cycling '74 (the company that makes Max) around 2010.

Functionally, programming in Max For Live or Max is exactly the same with the difference being that devices you build in Max For Live will only run inside of Ableton Live whereas building in Max standalone lets you create independent applications.

Max For Live is included with Ableton Live Suite (or as a separate add-on to Live Standard) but you'll have to buy Max as a separate application if you want to use it standalone.

For most people, Max For Live is all you need and the rest of this guide will focus exclusively on that.

How To Learn Max

The basics of Max are relatively simple to learn. You can probably learn everything you need to build basic patches in a few hours.

The complexity comes with learning all the different objects that are available and what they do, as well as figuring out complex logic and DSP algorithms.

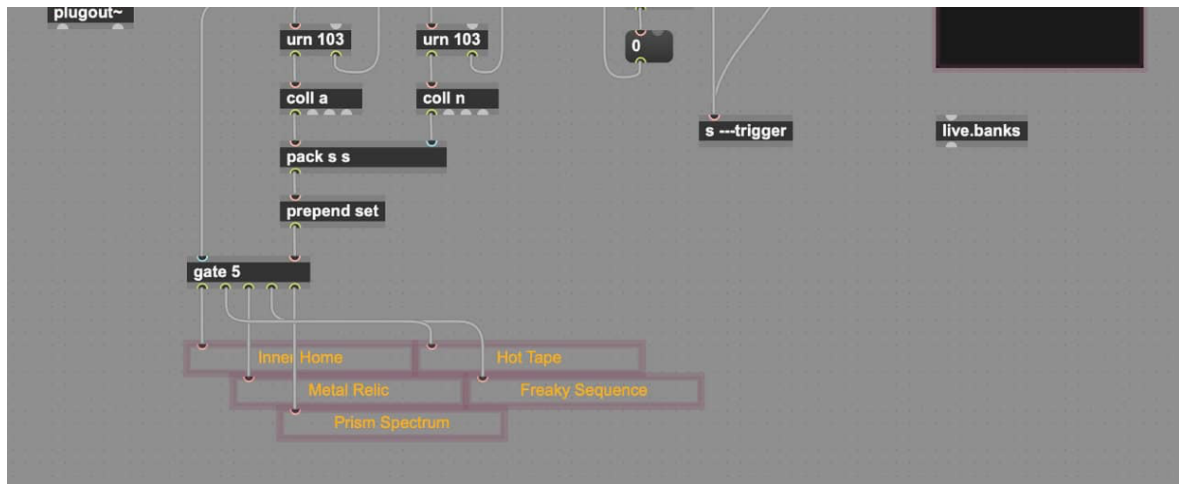
While the basics of Max won't take you long to get the hang of, building a mastery of everything you can do with Max is something that just takes time and dedicated practice to get better at.

Learn As You Go

You don't ever need to learn everything that Max can do and the way most people learn, once they've mastered the basics, is simply by learning bits and pieces as needed.

That's one of the joys of Max; it can very quickly be applied to the problem at hand – you don't need to learn everything up front, you can learn something new as and when you need to.





Max doesn't have to be complex. One of my most used devices (seen above) is only a handful of objects and took not much longer than 30min to make.

Learn From Others

Another beauty of Max is that it is incredibly open.

You can open devices made by anyone else and take a look inside to see how they function. This can be a great way to learn, you don't even need to understand the whole device. Perhaps all you're after is the filter in a complex polyphonic synth. Open the patch, pull out the filter component and learn how it's made.

The Max community is also very open, with an ethos of sharing and helping each other which makes learning Max a very enjoyable experience.

Learning The Basics

Now that you know how to approach learning Max, let's get those all-important basics down.

As I mentioned, in a few hours you can probably cover the fundamentals. In this section, I've collected a handful of excellent resources that will get you up to speed on the basics.

A Beginner's Guide To Building Max For Live Devices

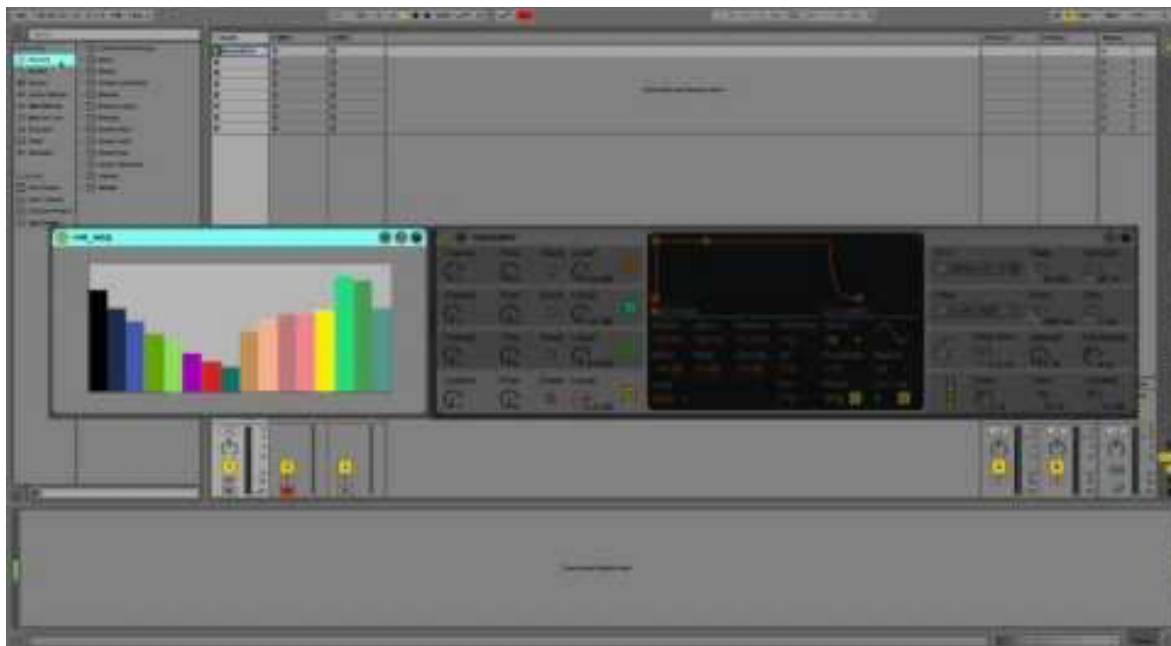
First up is one from yours truly.

This is a workshop I gave on the basics of Max For Live for the Ableton User Group, Cape Town. I approach things very slowly and deliberately, so if you want something super basic to get started then check this one out.



Programming In Max For Live

Next up is a collection of videos from Cycling '74, the company behind Max. This series is also a great intro to the fundamentals, walking you through building a simple audio effect, all the way up to your first polyphonic synthesizer.



[Access the full playlist on YouTube \(https://youtube.com/playlist?list=PLasl9l6VeCCrNLAoOiKibDqJc1rsjLSDi&si=0Jp6Oh3RNiqlIZ0i\)](https://youtube.com/playlist?list=PLasl9l6VeCCrNLAoOiKibDqJc1rsjLSDi&si=0Jp6Oh3RNiqlIZ0i)

Max For Live Beginner's Masterclass

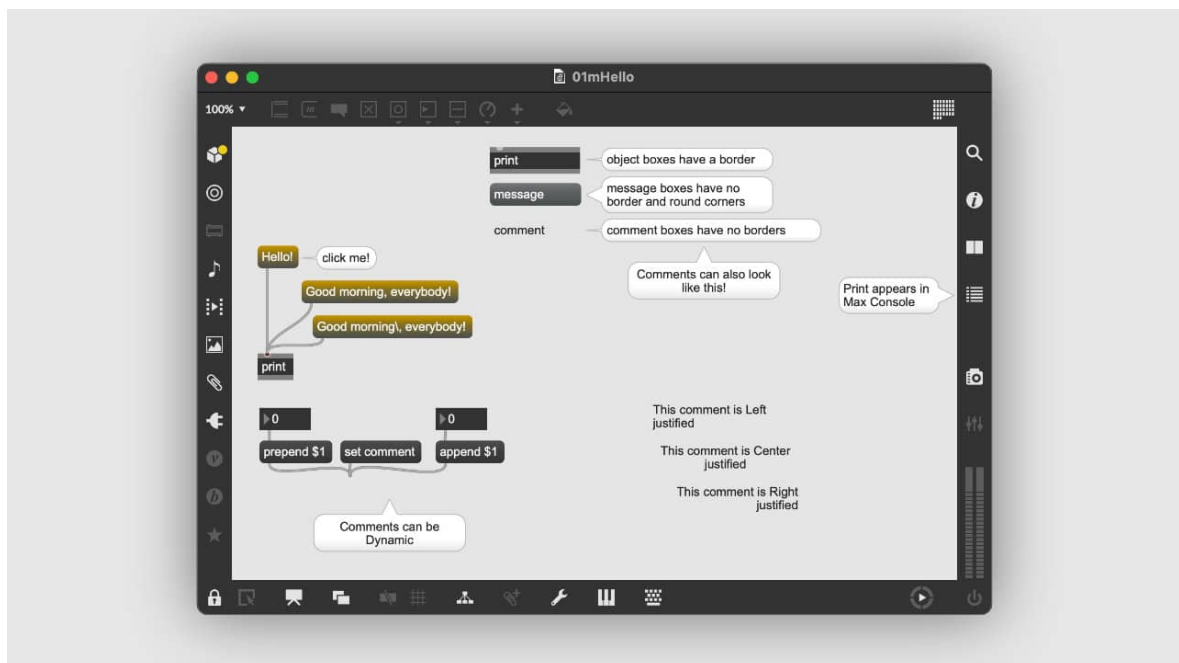
Learning Max is complicated! So it doesn't hurt to cover the basics a few times. Here's another excellent beginner's walkthrough of building a simple audio effect in Max from fellow Certified Trainer Phelan Kane.



Learn Right Inside Of Max

Probably the best resource for learning Max is the tutorials built right into the software.

These are text-based tutorials that let you follow along at your own pace through each step. They also all contain companion example patches that you can open and explore to learn what's going on.



The introductory 'Hello World!' example Max patch from the first tutorial.

You'll access these tutorials directly within Max. Unfortunately, they're a bit hidden so, if you're a beginner diving into Max for the first time, you might not even know they're there!

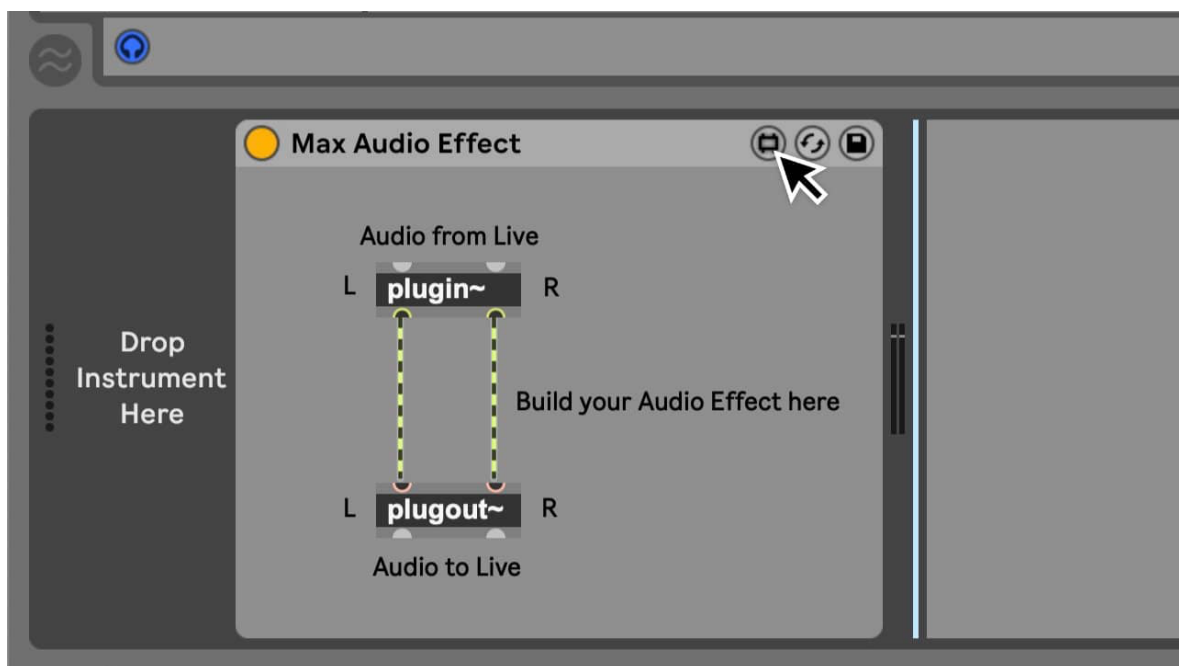
They also require you to know how to even get into the Max environment from within Live, which might not be obvious if you're new to Max!

I'll walk you through the steps to find the tutorials in the next section, but I'd suggest watching a few of the videos above to get a visual introduction to Max. You'll then be more equipped to open Max and find these tutorials.

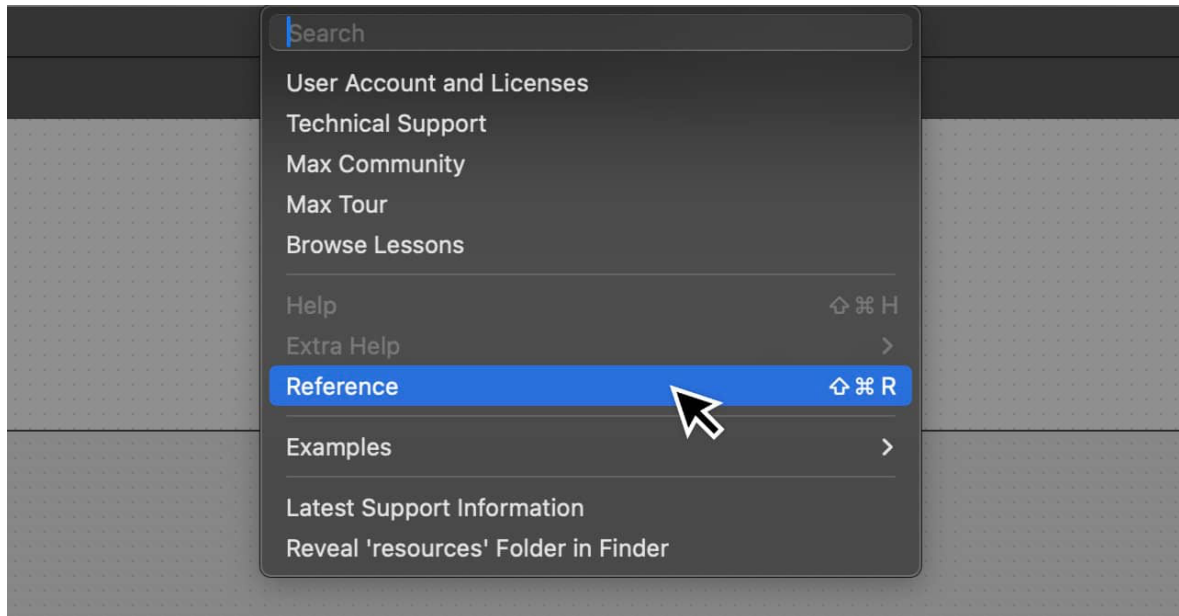
How To Find Max's Built-in Tutorials

To find the built-in tutorials, follow these steps:

Load up a blank Max For Live device inside of Live, then click the Edit button to open it in Max.



Once Max is open, head to the Help Menu. Confusingly, there is a section under Help called Browse Lessons – ignore this! I've found these lessons to not be particularly useful. Instead, head down to Reference.

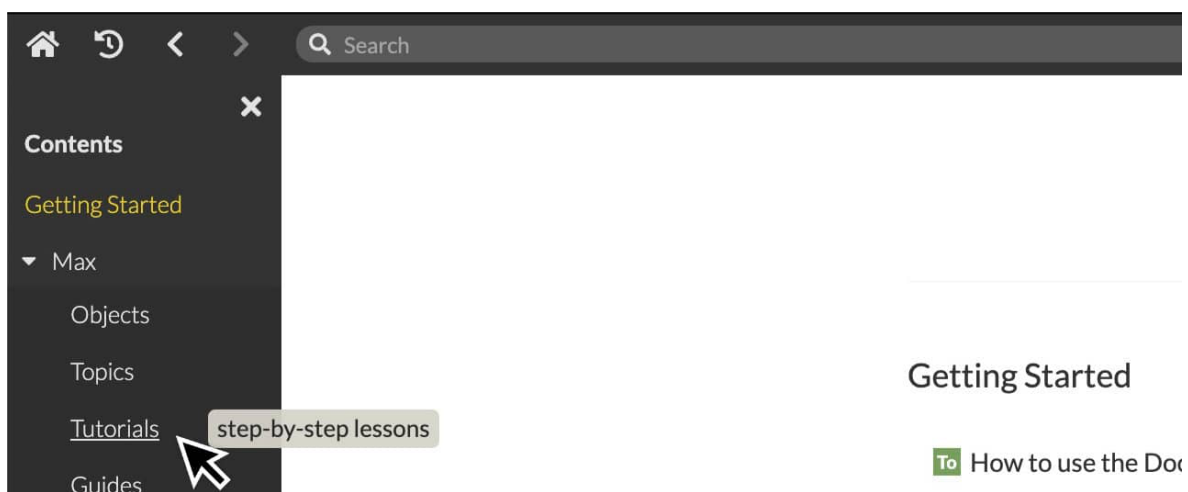


In the top left of the Reference window, you'll see a little 'Home' icon. Click this to go to the Max Documentation homepage.

You should now see a Contents menu down the left-hand side with categories for Max, MSP, Jitter, etc.

If you click the triangle toggle to open up each category you'll see sections labelled Tutorials. This is where you want to be!

From here you can navigate through the built-in tutorials and follow along at your own pace.



- ▶ MSP
- ▶ Jitter
- ▶ Gen

G How to use the Side

G Glossary of Commc

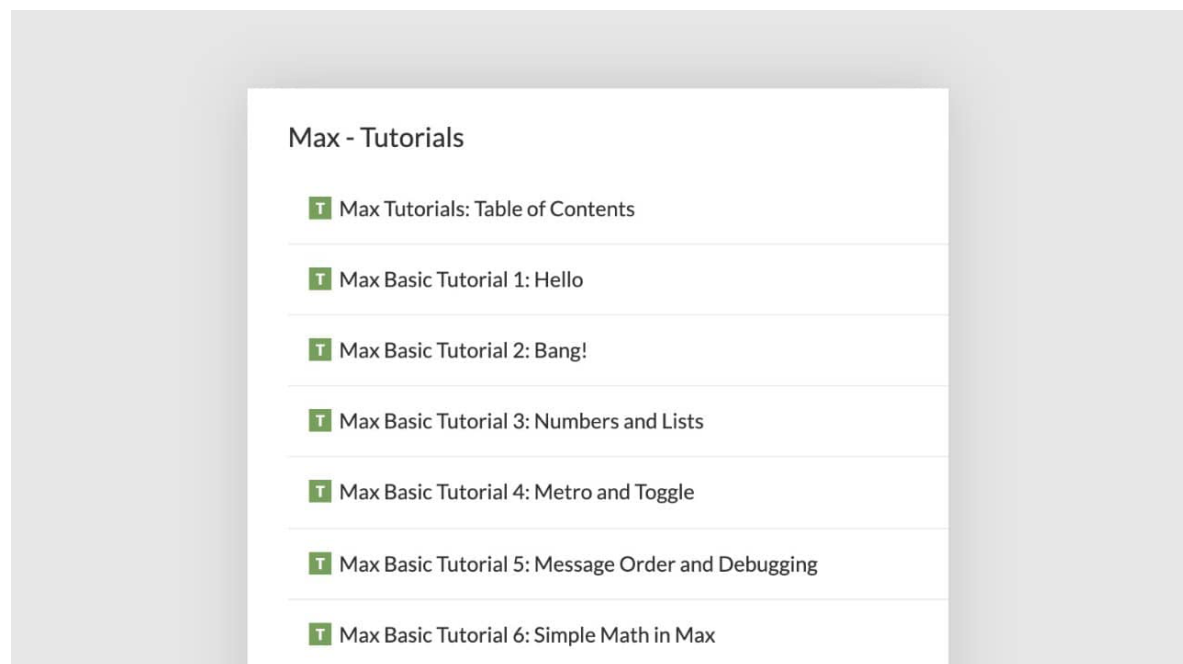
What Are Each Of These Categories?

Each category in the Reference relates to a different element of Max.

The **Max** category covers the core Max application. **You should start here.**

This has info on all the absolute basics you need to work with Max. You'll cover things like how to create and route data, set up interface controls, create MIDI, etc.

You don't need to go through all the tutorials in this section, but I'd recommend covering at least the *Max Basic* tutorials, and perhaps the *Max MIDI* tutorials as well.

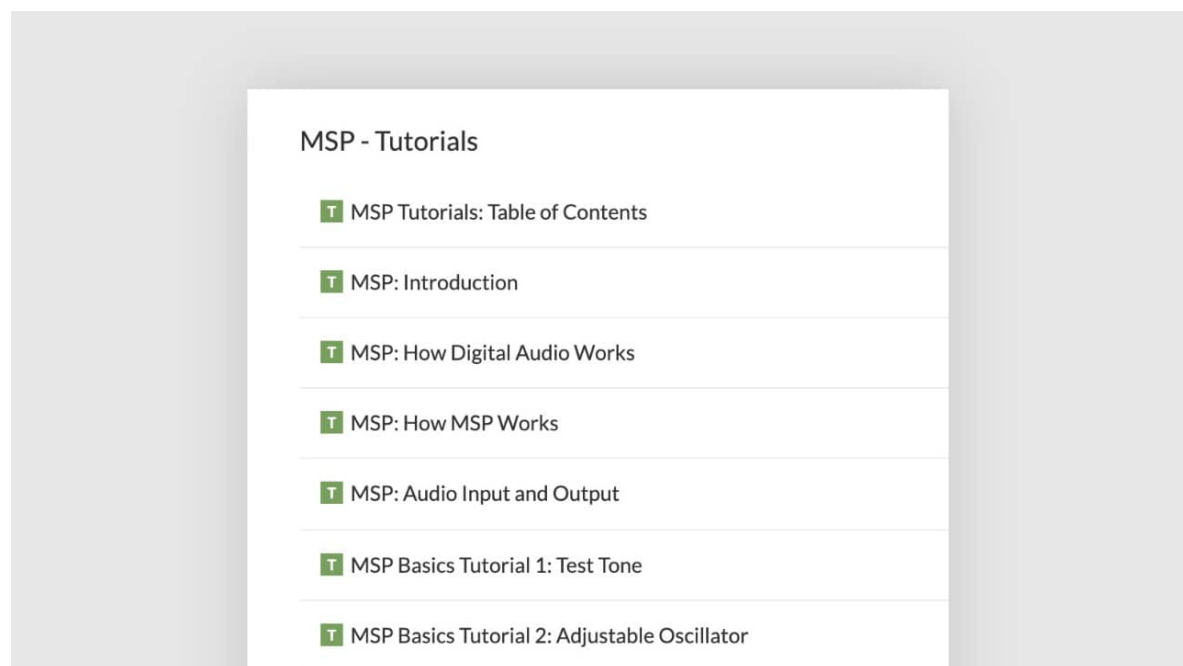


MIDI stands for Musical Instrument Digital Interface and deals with the audio representation

MSP stands for **Max Signal Processing** and deals with the audio processing side of Max. If you want to build patches that process audio (as opposed to only MIDI data) then you'll need to learn about MSP.

Once you've covered the Max basics, MSP is your next stop.

Again, you don't need to cover all of these, just the first handful of tutorials, but they're all incredibly useful if you're inclined to go through them.



The rest of the categories start to get into much more advanced stuff, so check these out only if or when you're interested in that particular area.

The two that are most immediately relevant will be **Max For Live**, which refers to elements of Max that specifically relate to using it inside of Live. There is also **Jitter**, which is the part of Max that lets you create visuals.

Additional Learning Resources

Everything covered above should give you a good start to learning Max, but there are plenty of resources you can use to go deeper.

Official Cycling '74 Resources

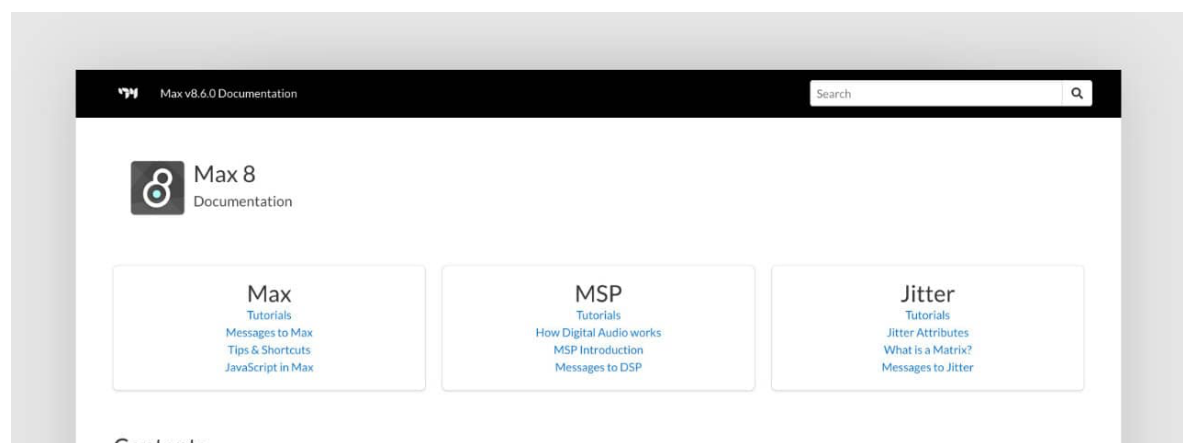
The Max documentation (everything you can access through the Reference we were just looking at) is incredibly thorough, so it's always a good place to go. You can also [access it online \(https://docs.cycling74.com/max8\)](https://docs.cycling74.com/max8).

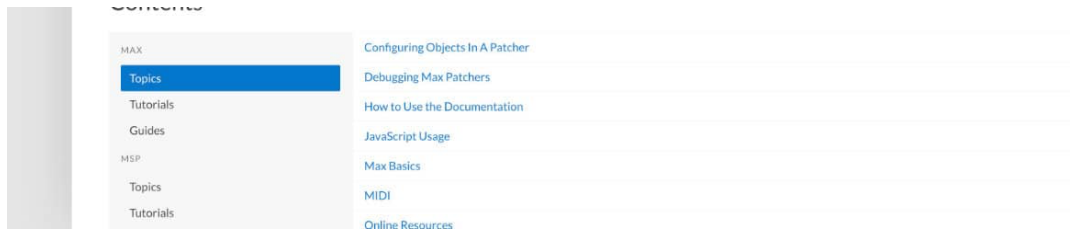
The [Max Learning Resources \(https://cycling74.com/resources\)](https://cycling74.com/resources) has some excellent content to check out.

The [Cycling '74 Forums \(https://cycling74.com/forums\)](https://cycling74.com/forums) are a good place to ask questions or to find older topics where people might have already asked and found answers to your questions.

There is tons of great Max learning content on YouTube.

[Cycling '74's channel \(https://www.youtube.com/@cycling74com/playlists\)](https://www.youtube.com/@cycling74com/playlists) has some great info, but there are 1000s of other Max creators that share their knowledge. Some of it can be pretty advanced, so it's not always the best place to go, but watching someone else build a Max patch can be really useful, even if you only pick up a few workflow tips or spot a cool little section of a patch you might reuse.

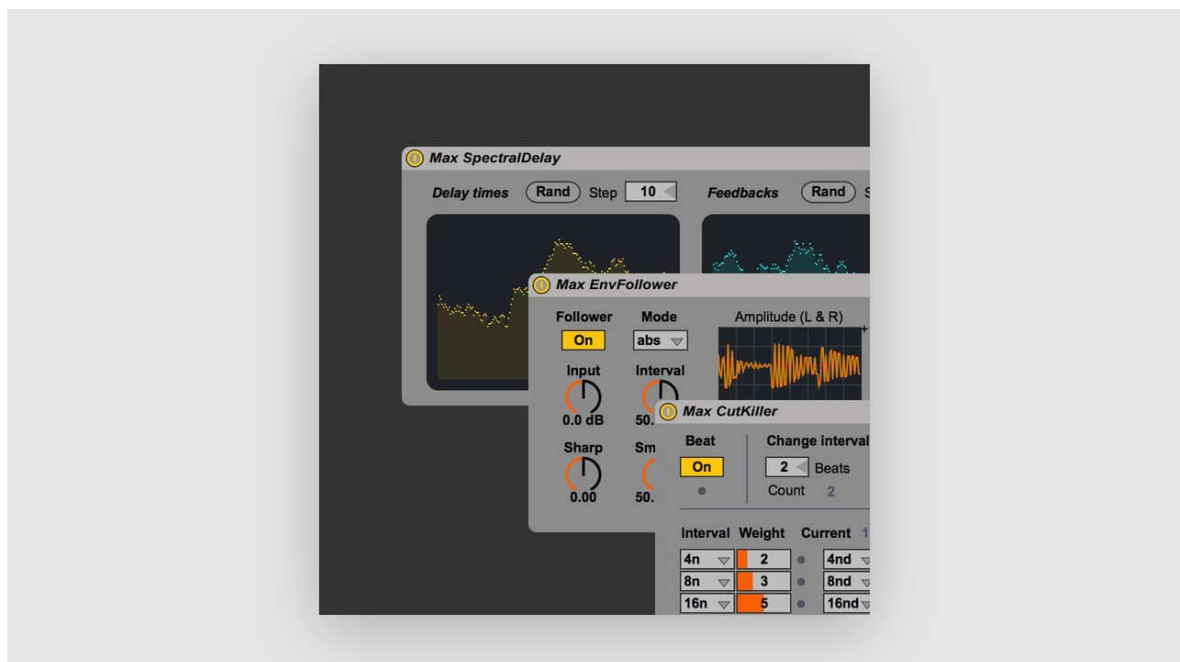




The official Max documentation is an excellent resource

Ableton Live Packs To Learn Max

There are some excellent Ableton Live Packs designed to help you learn Max.



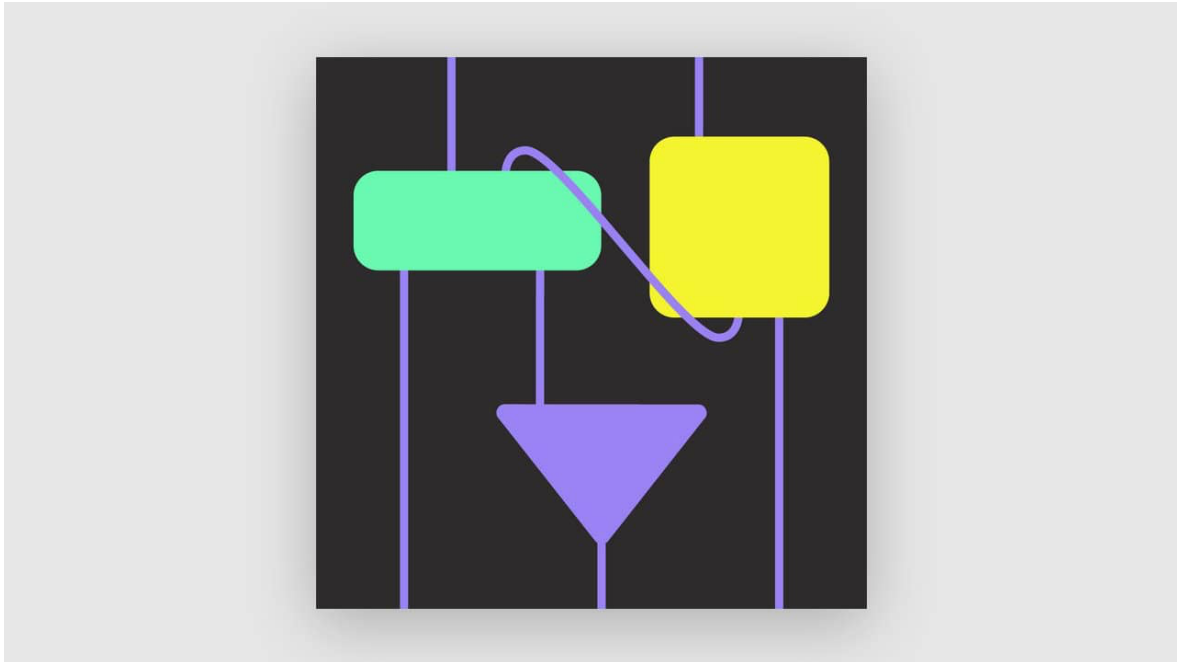
Max For Live Building Tools

The first is the Max For Live Building Tools

(<https://www.ableton.com/en/packs/max-live-building-tools/>).

This Pack contains hundreds of essential devices built in Max. The idea is that you can open each of them and see how they're made.

Each device is carefully annotated to explain how it's built. Once you've got the Max basics down, this is a great place to go to see some example Max patches and how to make them.



Building Max Devices

[Building Max Devices \(https://www.ableton.com/en/packs/building-max-devices/\)](https://www.ableton.com/en/packs/building-max-devices/) is another Pack designed to walk you through several example devices. It contains in-depth step-by-step lessons showing you how each device is made.

I found this one to be a little more on the advanced side, so spend a bit of time with the other resources in this post before checking this one out.

Learn From Others

Lastly, dissecting other's Max and Max For Live patches is a great way to

Lastly, dissecting other's max and max for live patches is a great way to learn. If you've got any Max For Live devices that you like, open them up and see how they're made.

Quite regularly, if I'm trying to build a device I've never built before, I'll head to maxforlive.com (<https://maxforlive.com/>) and find a device that's in the direction I'm heading and see how someone else has approached it.

Sometimes it's total *patch cable spaghetti*, but quite often I'm able to glean a useful little bit of a patch I can reuse.

Happy Patching!

Learning Max can be a daunting task, but it's incredibly rewarding in so many different ways, so it's worth putting in the time. I wish you the best of luck on your Max journey!

I'd love to hear from you if you've found this resource useful or (maybe more importantly) if you didn't!

My goal is to make the ultimate resource for people to learn how to build Max devices, so I want to ensure that the information here is clear and easy to follow.

Please let me know if it isn't. (<https://elphnt.io/contact/>)

I'd also love to hear if you have any other resources you found useful when learning Max so I can add them to the list.

Thanks and happy patching!

Share



Join The Making Music Newsletter

Email

Join

(htt	(htt
ps:/	ps:/
/ww	/ww
w.y	w.in
out	sta
ube	gra
.co	m.c
m/c/	om/
EL	elp
PH	hnt
NT)	x/)

(http://discovery.com/g/p/qhU/rMr/wR/Z)
Privacy Policy.

Store Policies

Support ELPHNT